

Multiple Choice

1. **Answer:** B

Options: A) 1.6 ns; B) 2.5 ns; C) 4.2 ns; D) 6.9 ns

Note: Correct option: B - 2.5 ns

2. **Answer:** B

Options: A) 2; B) 250; C) 5,000; D) 10,000

Note: Correct option: B - 250

3. **Answer:** D

Options: A) Specific heat; B) Thermal conductivity; C) Electrical resistivity; D) Hall coefficient

Note: Correct option: D - Hall coefficient

4. **Answer:** C

Options: A) 1.2×10^9 m/s; B) 3.0×10^8 m/s; C) 1.5×10^8 m/s; D) 1.0×10^8 m/s

Note: Correct option: C - 1.5×10^8 m/s

5. **Answer:** A

Options: A) Electron; B) Meson; C) Photon; D) Boson

Note: Correct option: A - Electron

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'gamma rays'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'four hydrogen atoms and one helium atom'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '250'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** $n = 3, l = 1$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $1/3$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key